

Sequential Coordination: How Language Enables Complex Innovation Discovery

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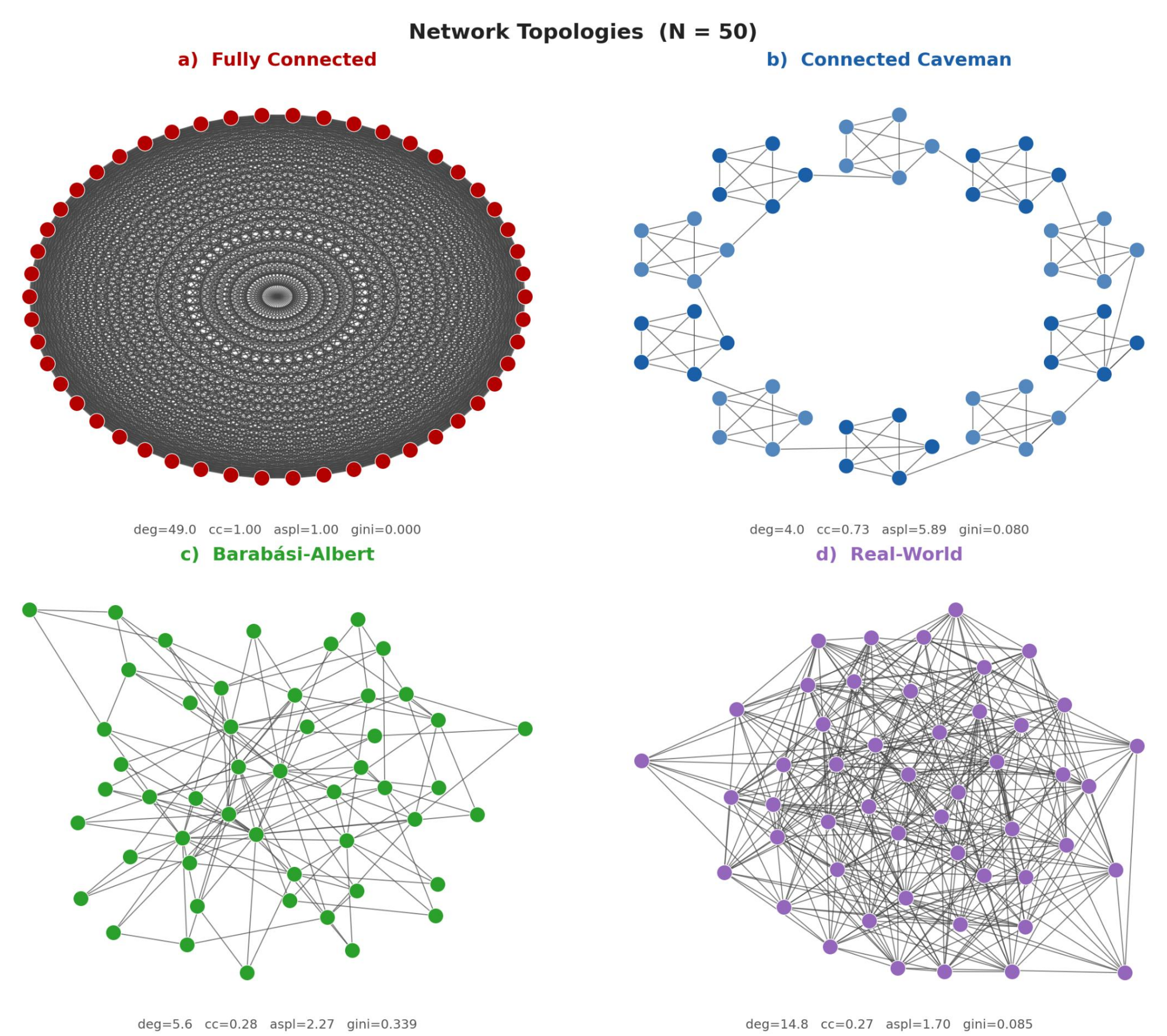
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Motivation

Dense networks promote linguistic convergence but may suppress the diversity needed for exploration. Sparse networks preserve diversity but fragment shared conventions. Can the **same network** support both language emergence and collective innovation?

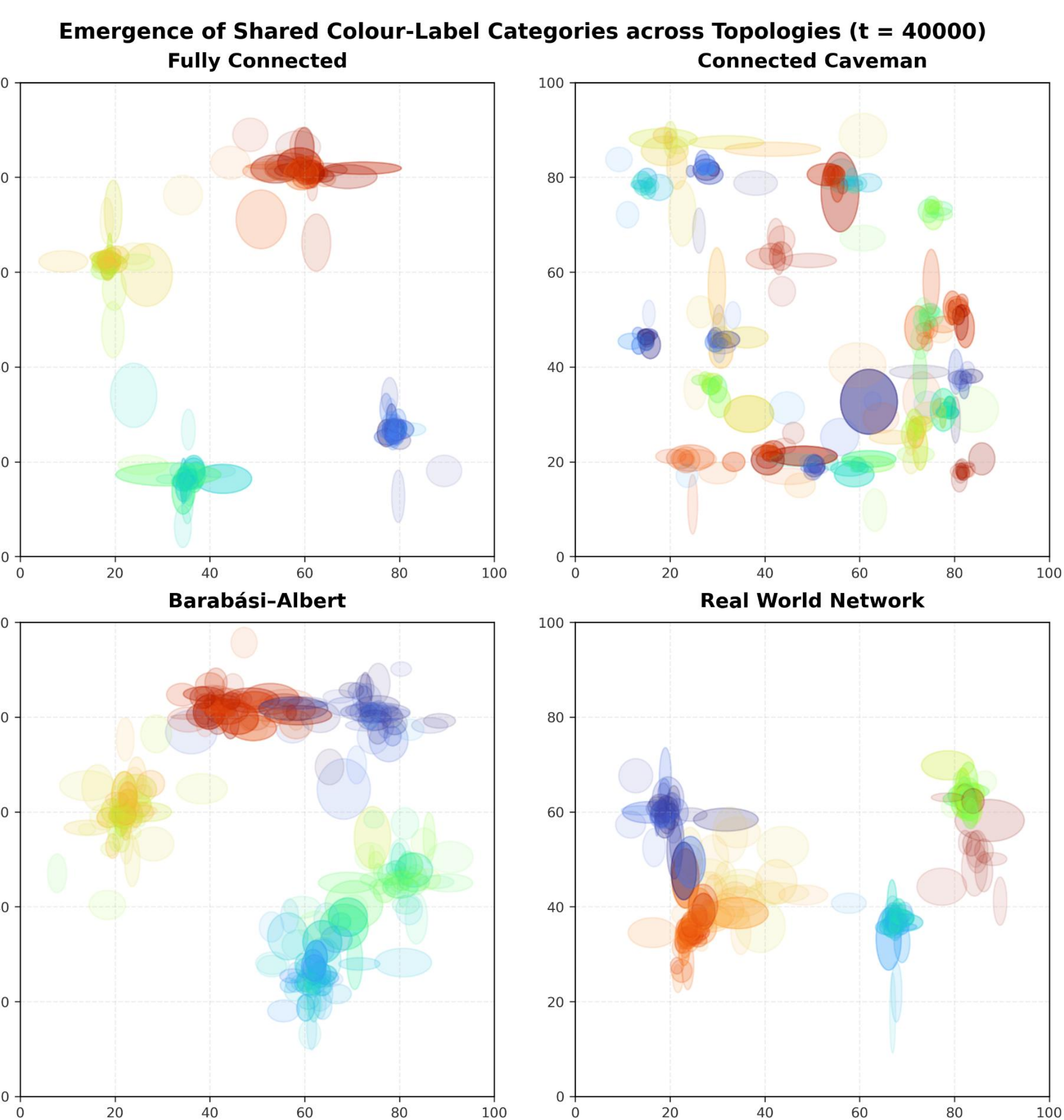
Four Network Topologies

The following setups are used to investigate the effects of information speed, local cohesion, and degree of heterogeneity, as well as a real-world scenario.



Study 1: Color Naming Game

In Sarkaboudi & Falandays(2025) we showed that all four topologies converge to **similar communicative success** (≈ 0.80), yet produce structurally different color-label category systems



If network structure shapes language form but not function, does the form of language still matter for what populations can achieve together? Study 2 tests this directly.

Study 2: Potions Game

Overview of the Game

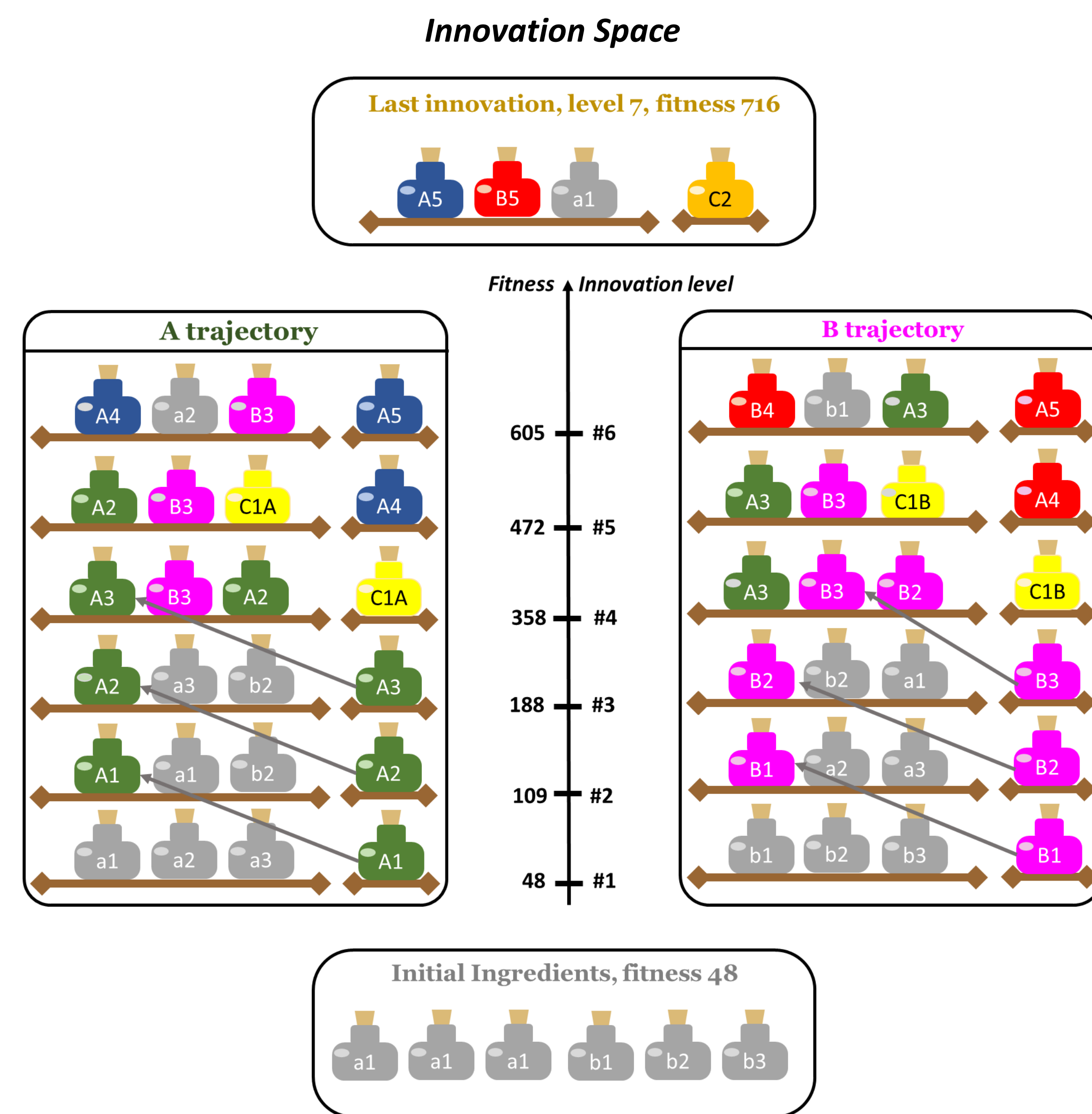
The game involves combining three potions to discover potions from higher tiers. In each iteration, a speaker and a listener are randomly selected. The agents function as Bayesian learners, maintaining Gaussian mixture models over a potion-label association matrix. Agents are required to coordinate on the labels while simultaneously seeking to innovate.

Speaker Act

Selects and labels (one or two) potion from their repertoire using a discriminative GMM: the category that most selectively identifies the target ingredient emits a label + Gaussian noise.

Listener Act

The listener hears the label, figures out which potion or potions are needed, and then chooses the remaining potions from their own collection using fitness-weighted sampling to prepare the three ingredients.

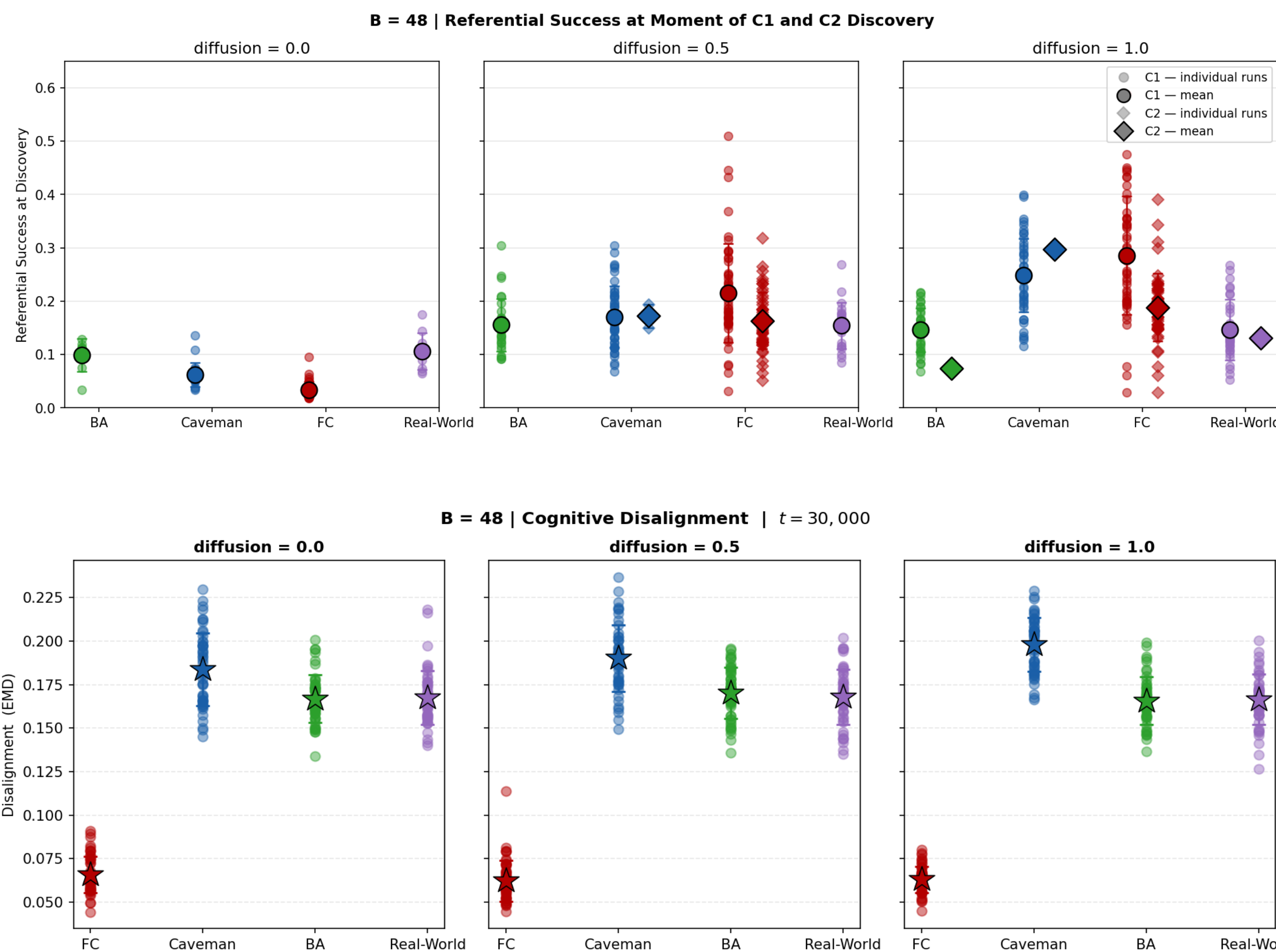
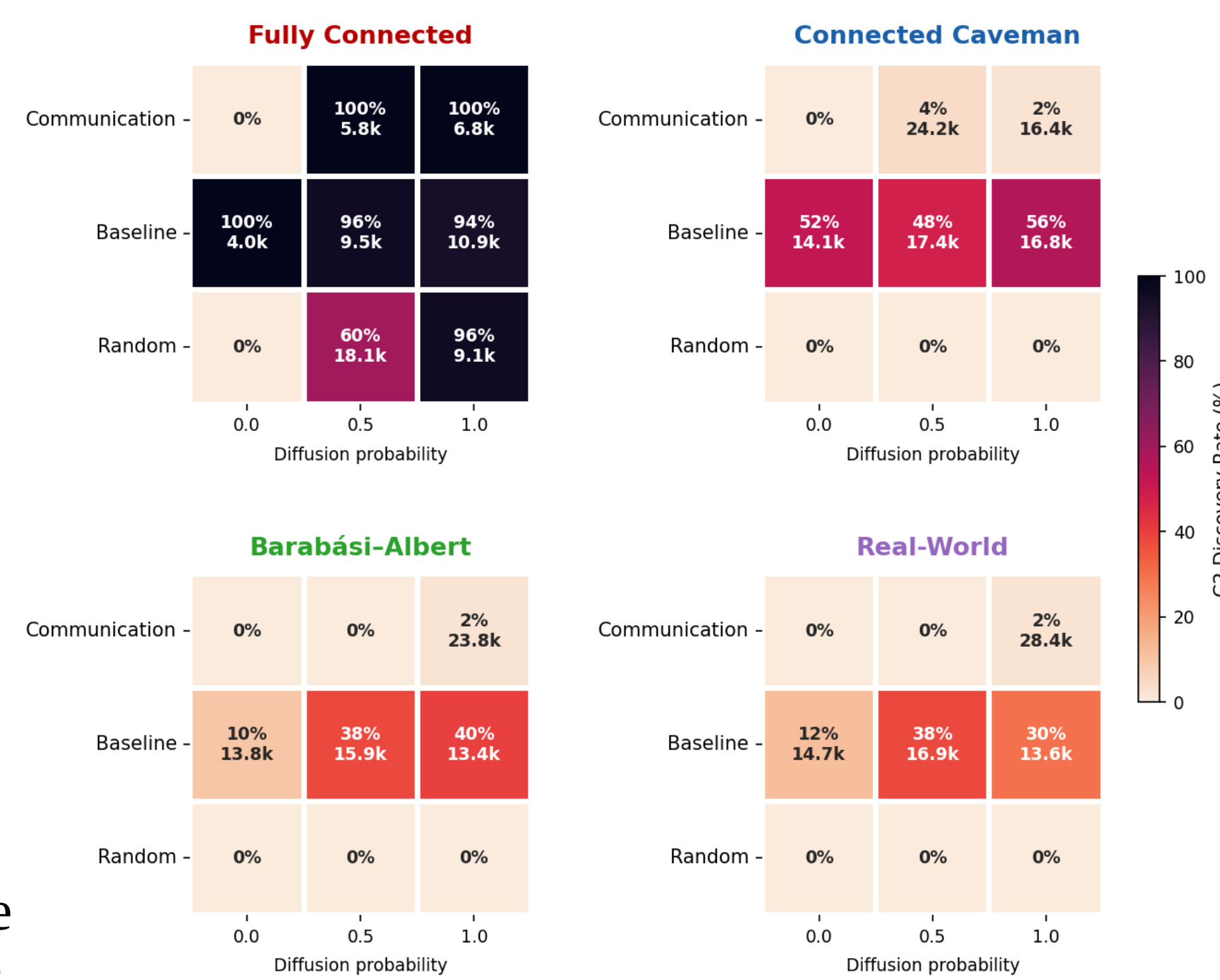


Results

We tested the game in three setups:

- Random:** Both agents choose randomly, without considering fitness or using language.
- Baseline:** Both agents select based on fitness-weighted sampling, without using language.
- Communication:** The listener chooses based on fitness, while the speaker's potion is communicated through language. The speaker labels the potion, and the listener tries to decode it.

B = 48 — C2 Discovery Rate (%) and Mean Steps



References

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Key takeaways

1 Network topology is the main factor in both studies. Dense networks help languages come together and spread what is needed, while sparse networks keep local diversity but can slow down innovation.

2 Linguistic imprecision can be useful. When references fail, they introduce unexpected elements that precise choices would miss. This process helps make new discoveries in C2.

3 Dividing roles helps resolve the exploration-exploitation trade-off in a single interaction. The listener focuses on using information, while the speaker's noise introduces new possibilities.

Linguistic imprecision is not just a cost of communication. It can actually help us make discoveries that strict accuracy cannot.